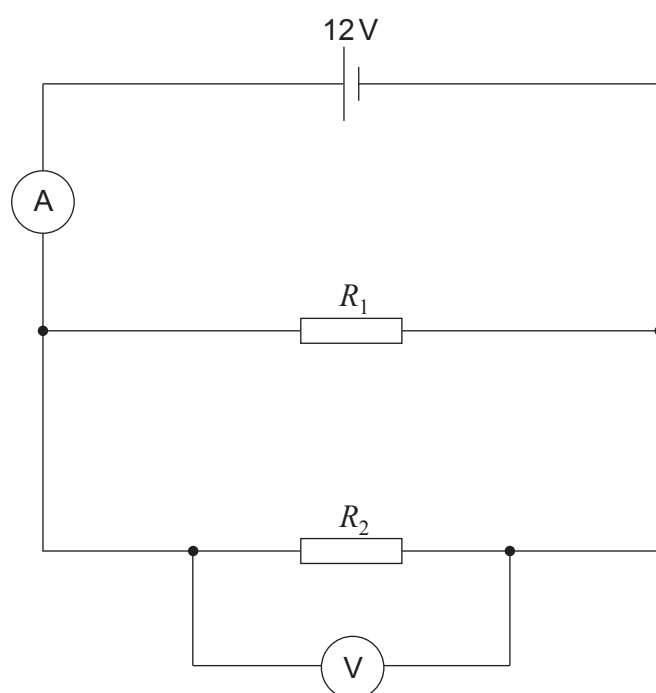


3. The circuit shown below contains two identical resistors connected in parallel. Each resistor has a resistance of 10Ω .



- (a) (i) Use the equation:

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2}$$

to calculate the total resistance of the circuit.

[2]

total resistance = Ω

- (ii) Use an equation from page 2 to calculate the current reading on the ammeter. [2]

current = A



(iii) State the voltage across R_2 .

[1]

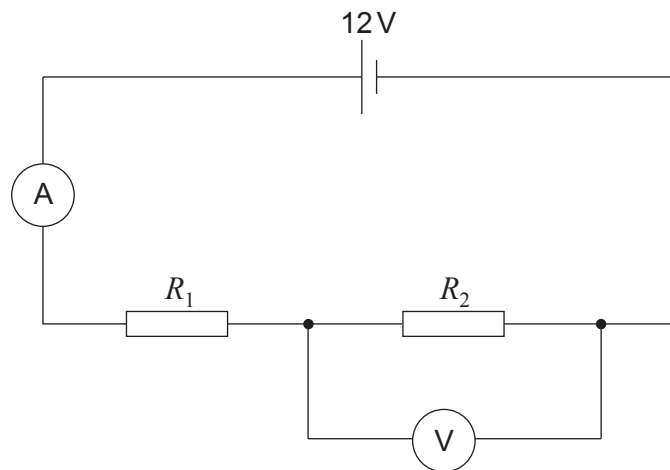
voltage = V

(iv) Calculate the current through R_2 .

[2]

current = A

(b) The same resistors are now connected in series.



State how the following values compare with the parallel circuit in part (a).

(i) The total resistance of the circuit.

[1]

.....

(ii) The current reading on the ammeter.

[1]

.....

(iii) I. The voltage across R_2 .

[1]

.....

II. Give a reason for your answer.

[1]

.....

.....

